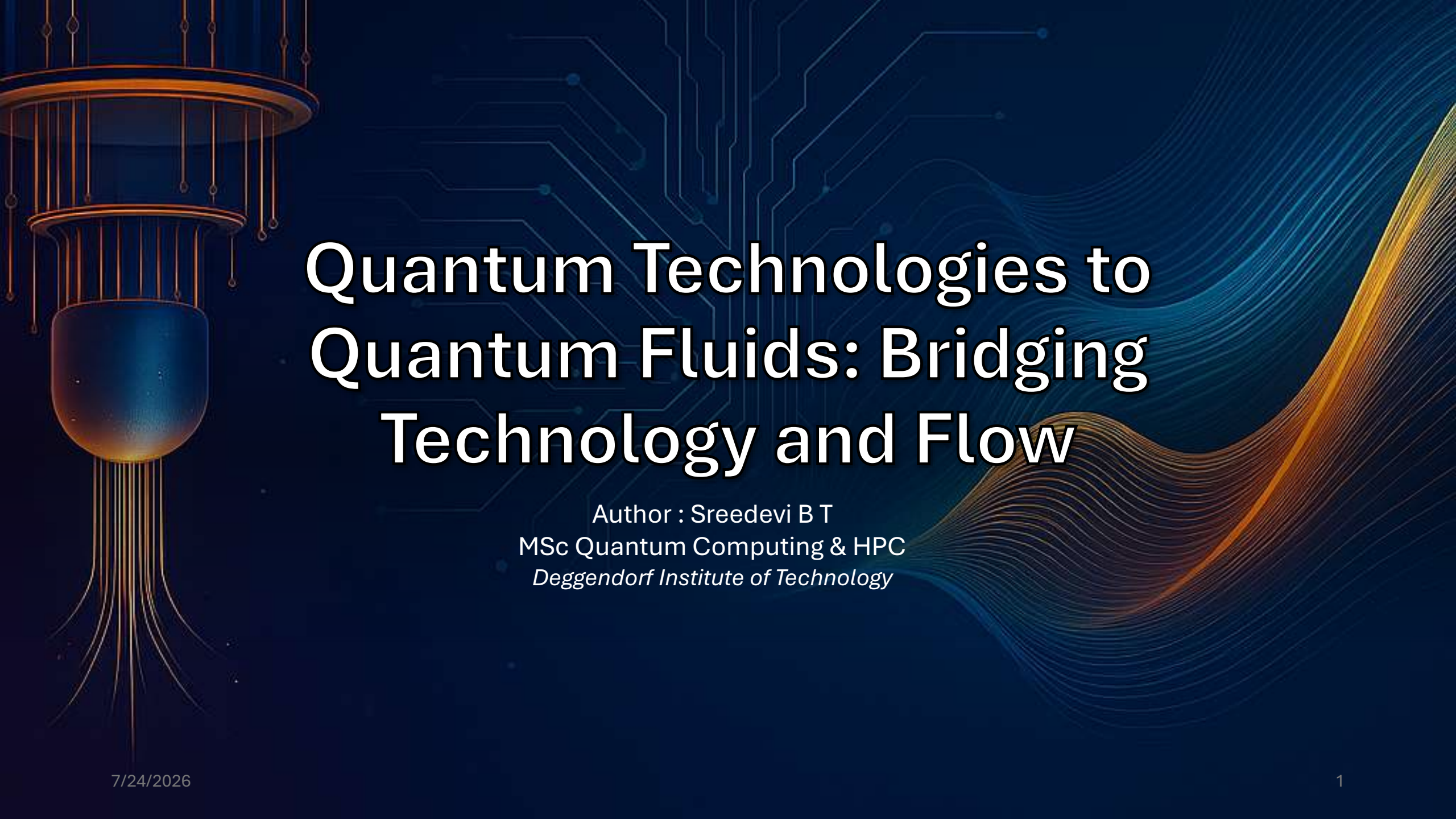


The background is a dark blue gradient. On the left, there is a stylized illustration of a hanging lamp or chandelier with multiple tiers and thin, vertical rods. On the right, there are flowing, wavy lines in shades of blue and orange, resembling liquid or energy. The overall aesthetic is futuristic and scientific.

Quantum Technologies to Quantum Fluids: Bridging Technology and Flow

Author : Sreedevi B T
MSc Quantum Computing & HPC
Deggendorf Institute of Technology



Short Bio

2024 ~ now:

- M.Sc. in Quantum Computing & HPC, Deggendorf Institute of Technology, Germany
- CTO & Co-founder – Quantum-AI Energy Copilot Startup (*Powerwise*)
- Industrial Collaboration with Eviden on Quantum-Inspired Optimization
- Quantum Lattice Boltzmann Method for Pollution Modeling

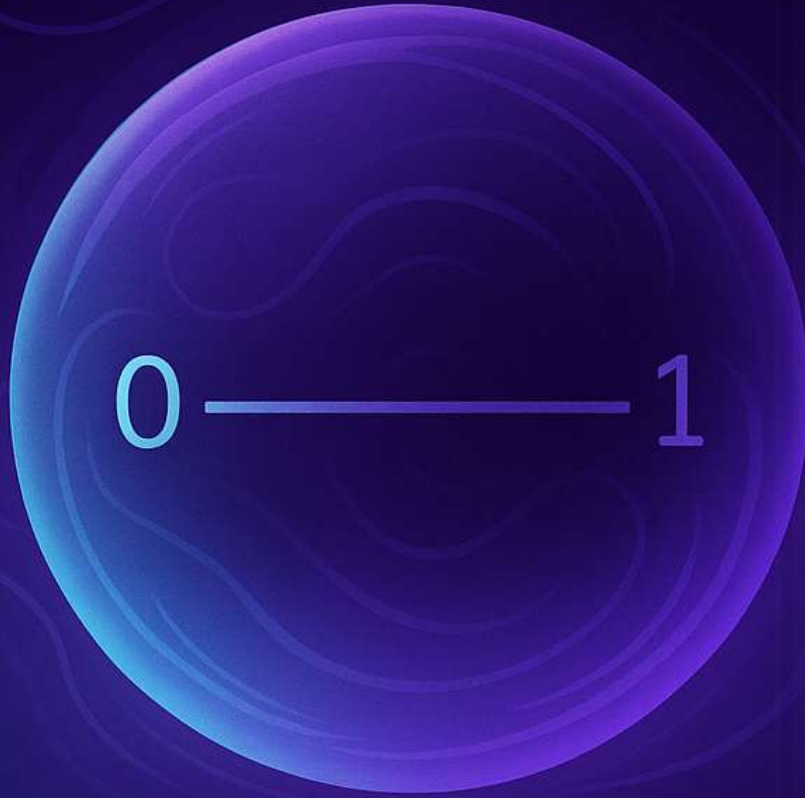
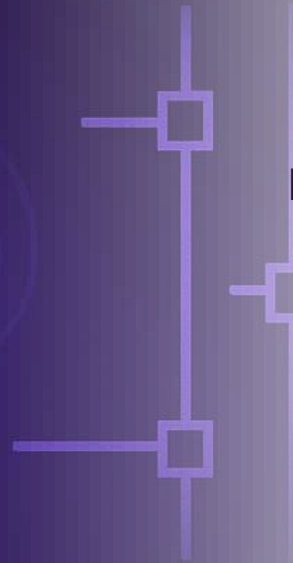
2020 ~ 2022:

- M.Sc. in Physics, Central University of Karnataka, India
- Thesis: Majorana Dark Matter in a Gauged Le - Li Model (MATLAB)

2015 ~ 2018: B.Sc. in Physics, Kerala University, India

Quantum Computing

- Quantum bits
 - Unlike classical bits (0 or 1), qubits can be in a superposition of 0 and 1.
- Entanglement
 - Links qubits across space for richer information handling.
- Quantum gates and circuits
 - Computation is done via quantum circuits using quantum gates.





Quantum Computing : Applications

- Quantum Chemistry
 - Simulating molecular structures and chemical reactions
- Healthcare
 - Drug discovery, protein folding, disease modeling
- Climate science and Computational fluid dynamics(CFD)
 - Modeling turbulence, large scale simulations
- Logistics and Scheduling
 - Vehicle routing, multi-objective optimization

CFD: Workflow and Techniques

- **Workflow**

- Pre-processing
- Solver Setup
- Simulation
- Post-processing

- **Traditional techniques:**

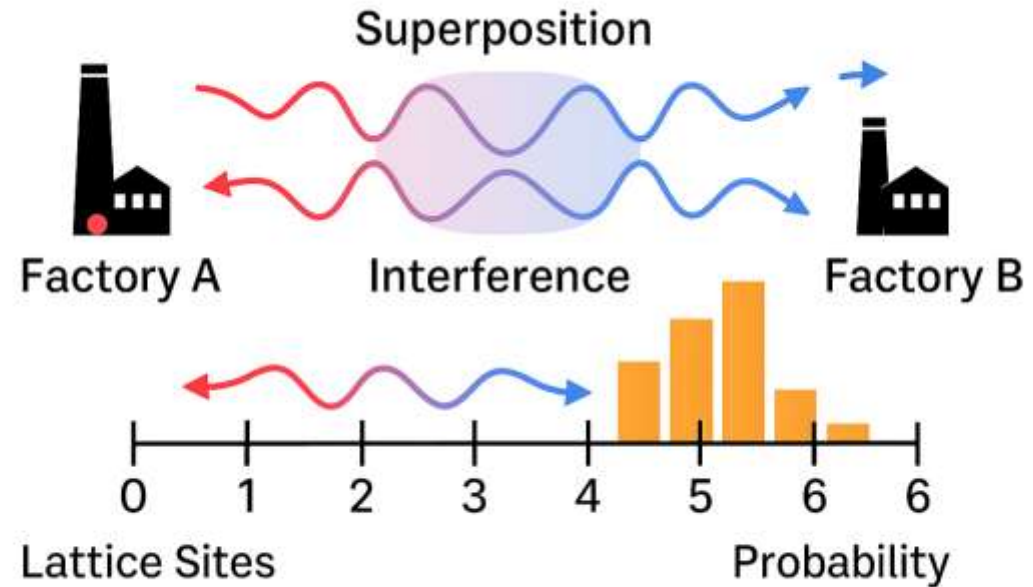
- - Finite Volume Method (FVM)
 - Lattice Boltzmann Method (LBM)
 - Finite Element Method (FEM)

Our Expertise in Quantum Computing: QLBM for Pollution Modeling

- Gaussian-Hill advection diffusion animations - Simulation in Quantum System v.s. Analytical Solution: 1D, 2D, 3D (left to right)
- A mesoscopic approach where the macroscopic dynamics are approximated by interactions between fictitious particles on a regular lattice, significantly interesting as it highlights quantum nativity.
- Published in top computer science journal “**ACM Transactions on Quantum Computing**”: *Utilizing classical programming principles in the Intel Quantum SDK: implementation of quantum lattice Boltzmann method*, <https://doi.org/10.1145/3678185>

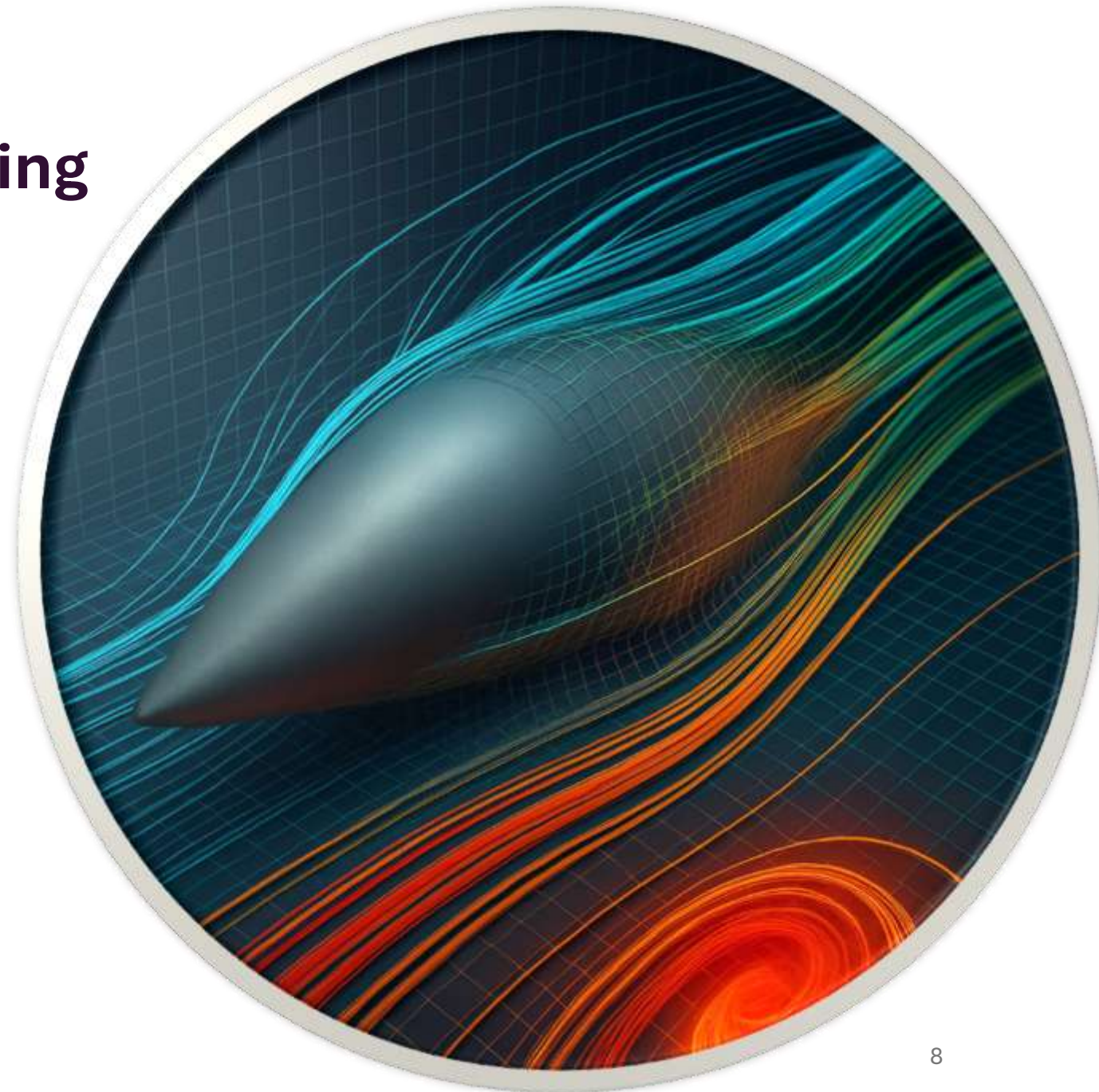
Simulating Pollution Spread Using 1D QLBM and LBM

- Focus: Simulating pollutant spread in a 1D spatial domain
- Study objective: Track how concentration of particles evolves over time from multiple sources under wind and turbulence.
- Importance: Predict and analyze air pollution from multiple sources in linear setups (e.g., tunnels, narrow streets)



Implementation Of Pollution Spread In Air Using D1Q3 LBM

- Advection: Modeled wind flow carrying pollutants.
- Diffusion: Represented atmospheric turbulence spreading the pollutants.
- Decay – Chemical or natural removal
- Source Interaction – overlapping plumes
- Multiple Sources: Pollutant release at specific positions in the domain.



Lattice Boltzmann Method : Core

- The Lattice Boltzmann Equation for a 1D advection – diffusion problem can be written as

$$\partial_t C + u \partial_x C = D \partial_{xx} C - \lambda C - k C^2$$

- **Equilibrium distribution function**

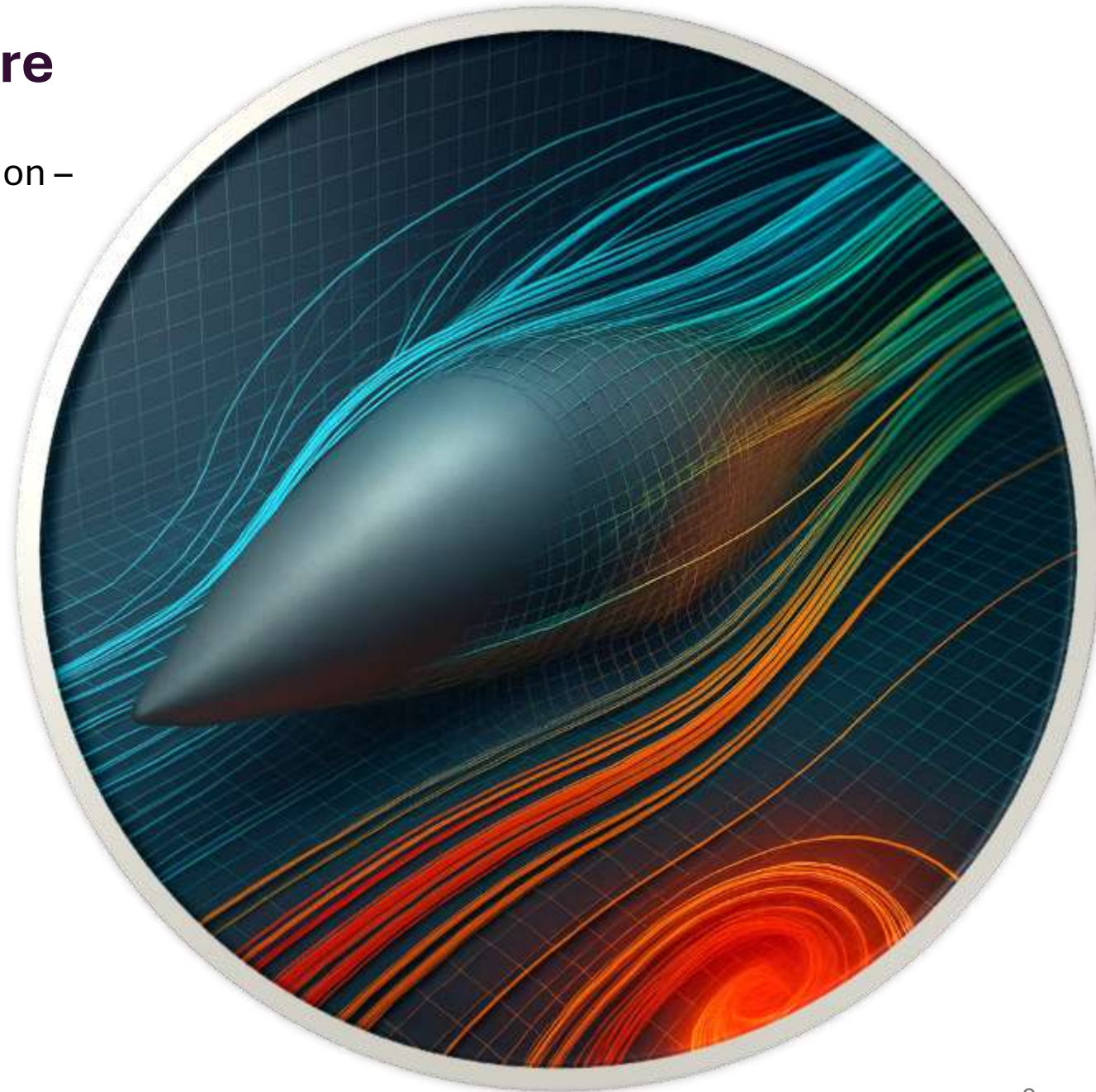
$$g_i^* = g_i - \frac{1}{\tau} (g_i - g_i^{eq}) + w_i R(C) \Delta t$$
$$R(C) = -\lambda C - k C^2$$

- **Collision operator with reaction term**

$$g_i^{eq} = w_i C \left(1 + \frac{e_i u}{c_s^2} \right)$$

- **Streaming step**

$$g_i(x + e_i \Delta x, t + \Delta t) = g_i^*(x, t)$$



- Discrete velocities: $e = \{-1, 0, +1\}$ (D1Q3)

- Weights: $w = \{1/6, 2/3, 1/6\}$

- Speed of sound: $c_s^2 = 1/3$

- Relaxation time:

$$\tau = 0.5 + \frac{D}{c_s^2}$$

- Macroscopic field:

$$C(x, t) = \sum_i g_i(x, t)$$

- Mach number constraint:

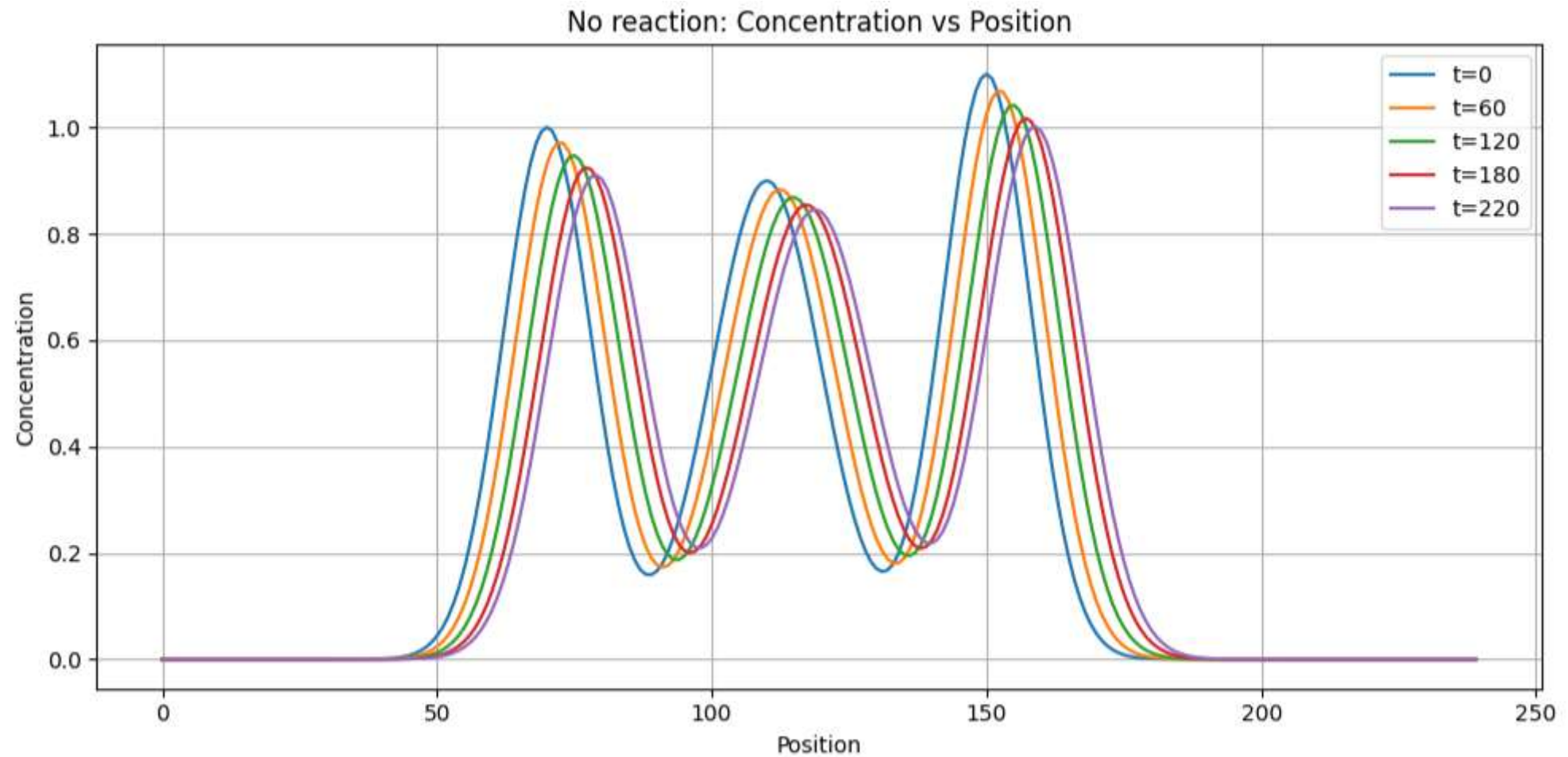
$$Ma = u/c_s \lesssim 0.3$$

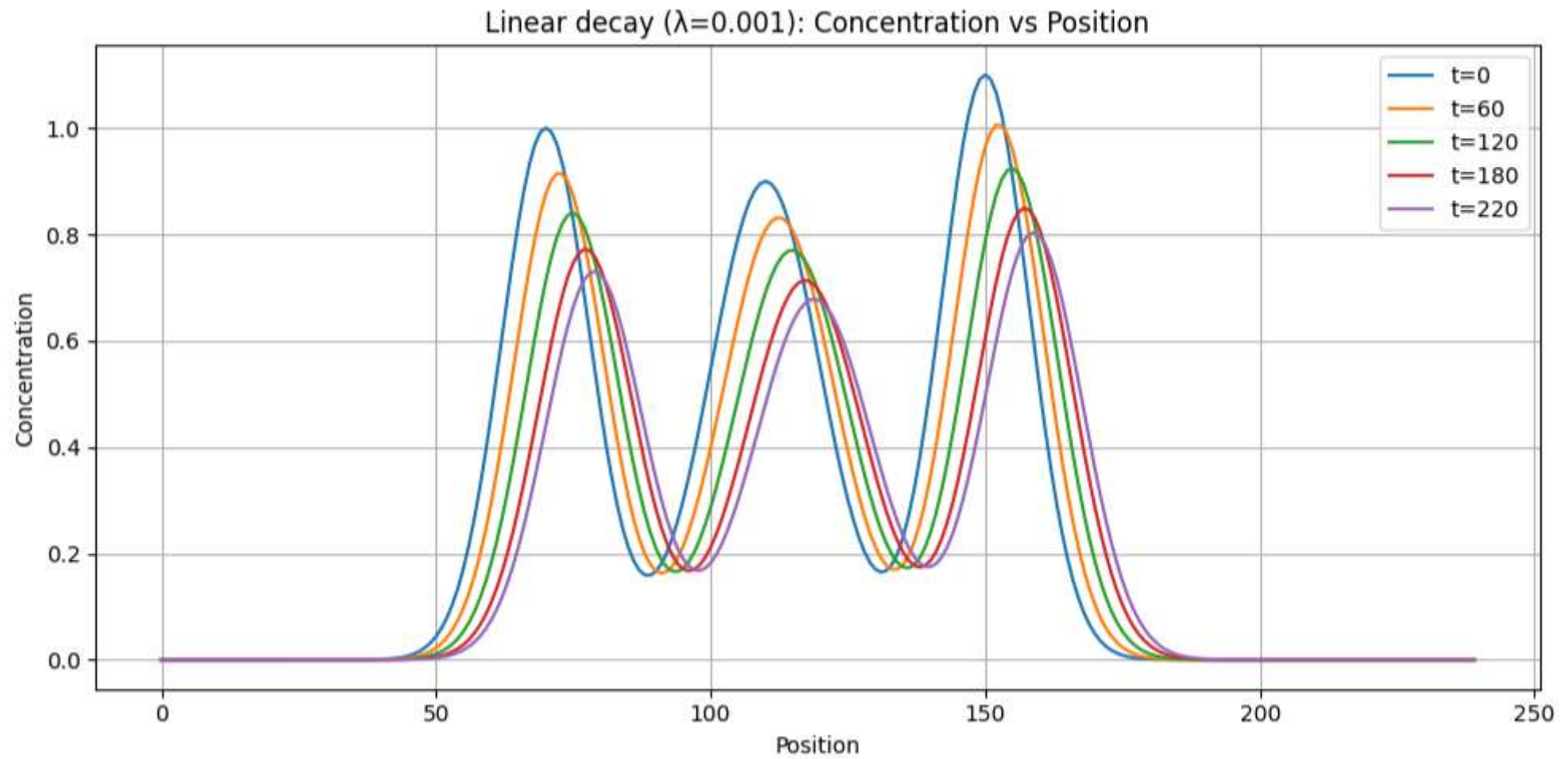


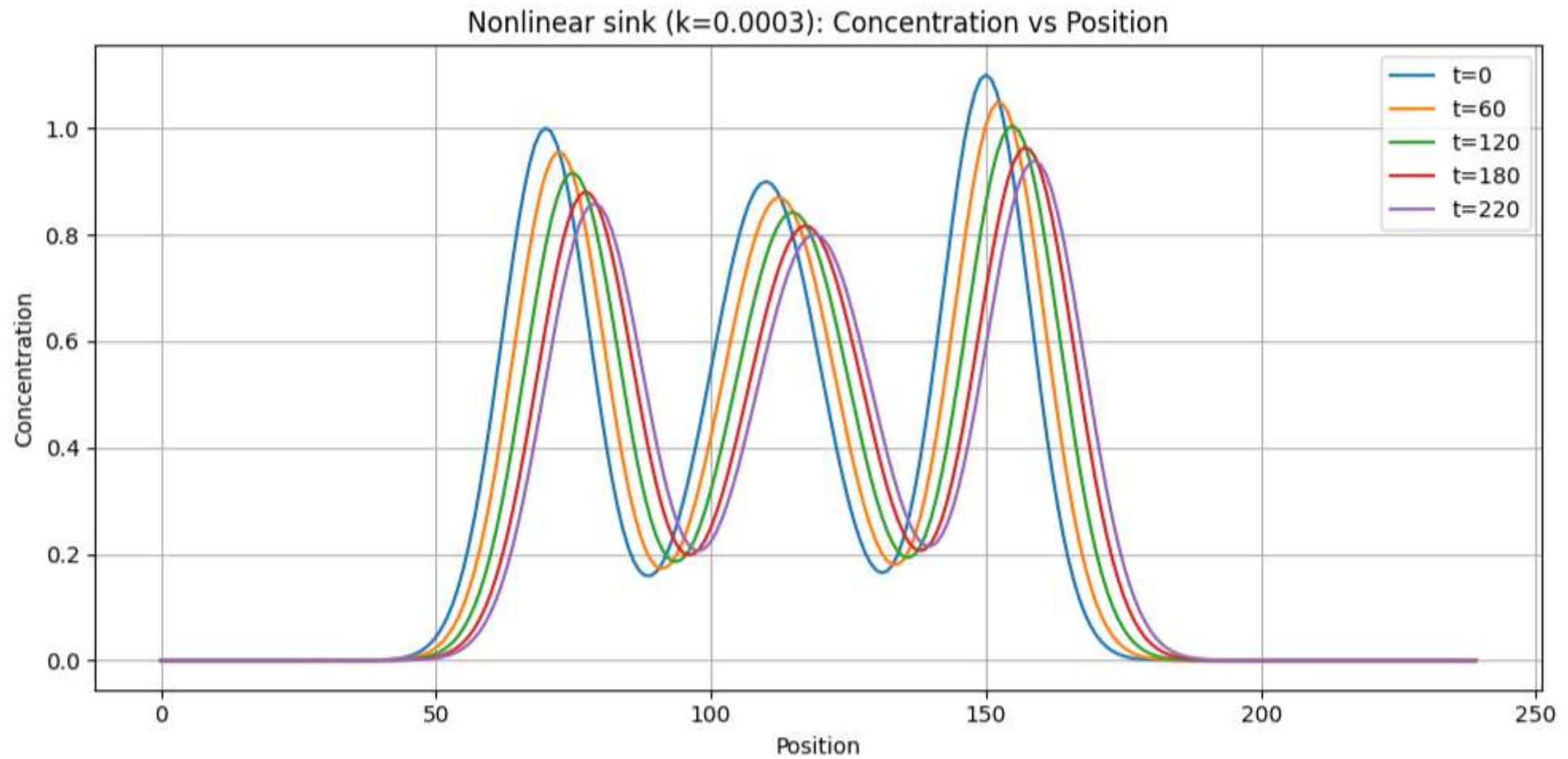
Conditions in our context

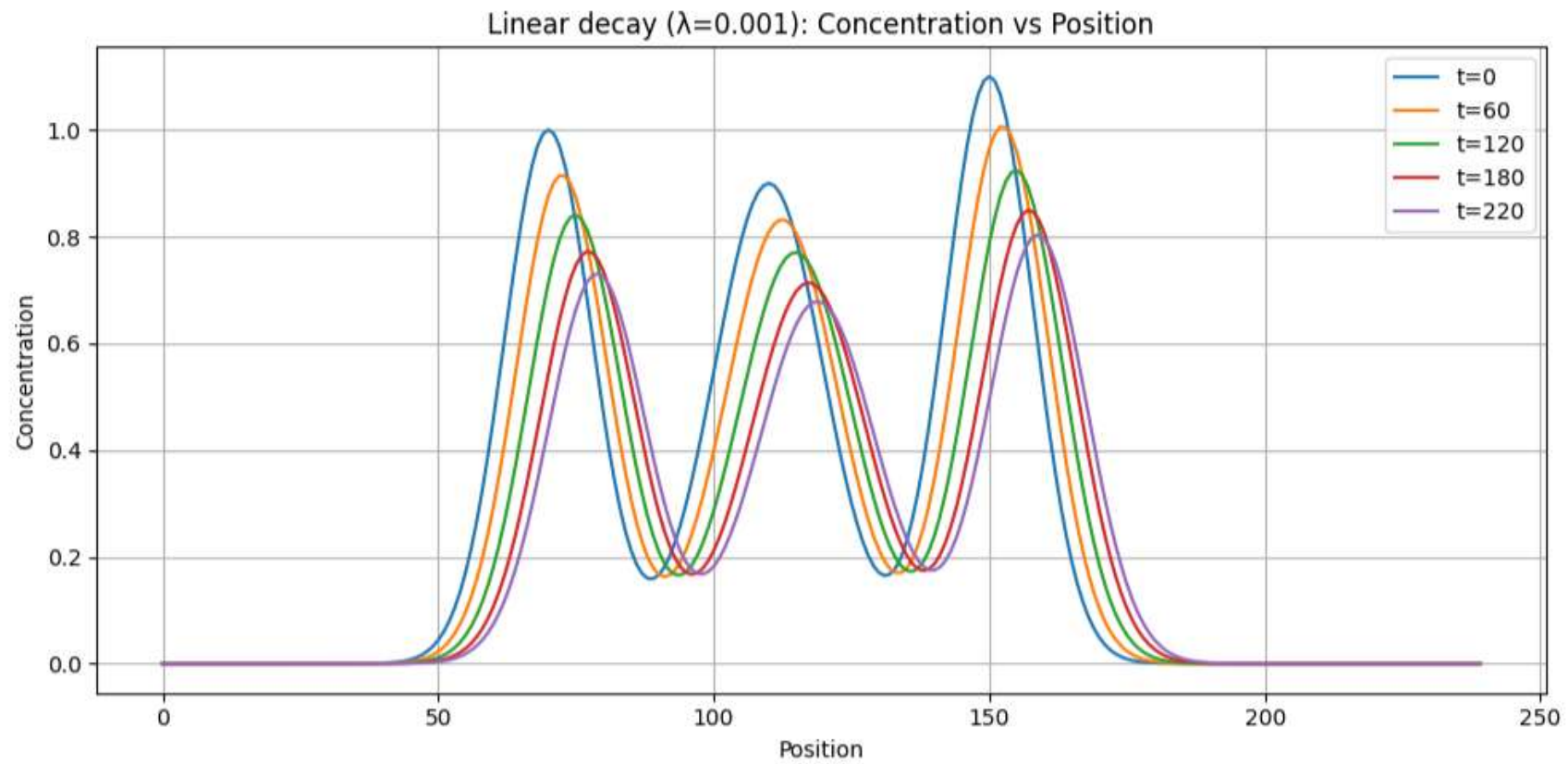
- Domain and Grid: A 1D domain with 240 spatial nodes ($x = 0$ to $x = 239$) and a time horizon of 220 time steps.
- Advection Velocity (u): Set to 0.04 to represent low wind speed and ensuring subsonic flow, $Mach = 0.07$.
- Diffusion Coefficient (D): Set to 0.03 to model moderate atmospheric turbulence.
- Sources: Pollutant sources are defined at positions $x = 70$, $x = 110$, and $x = 150$, each with gaussian concentration profiles of amplitude = 1.00, 0.90 and 1.10, width = 8.0, 10.0, 8.0.
- Boundary Conditions: Periodic boundaries are applied allowing the pollutants to reenter from the other, imitating a repeating spatial domain

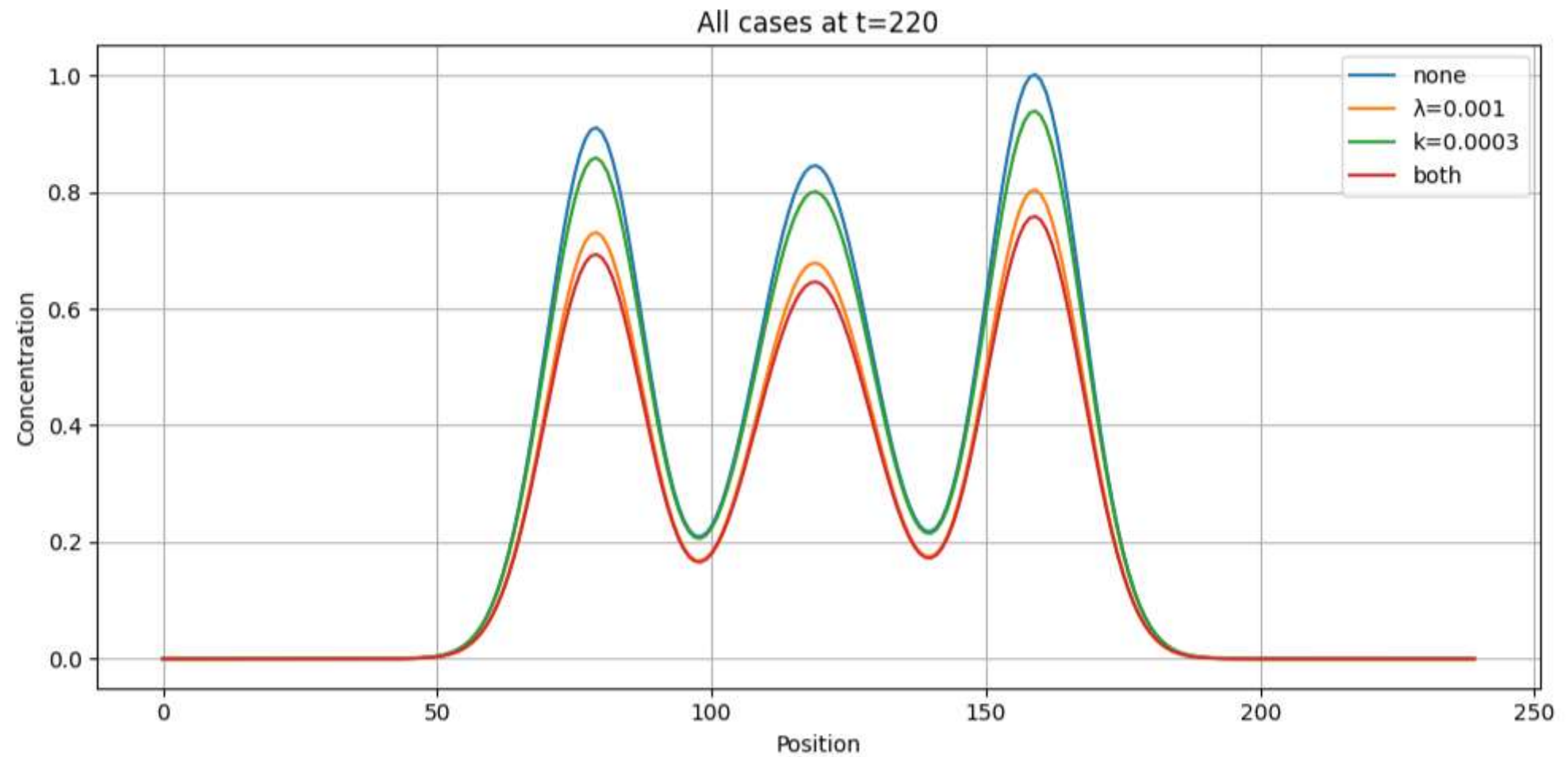
Results – Line plot











Implementation And Simulation Set Up Using Quantum Lattice Boltzmann Method(QLBM)

Qubit Encoding

- Use 8 position qubits
- Use 2 velocity qubit ($|00\rangle$ = rest, $|01\rangle$ = right, $|10\rangle$ =left , $|11\rangle$ =unused)
- Velocity weights= $2/3$, $1/6$, $1/6$

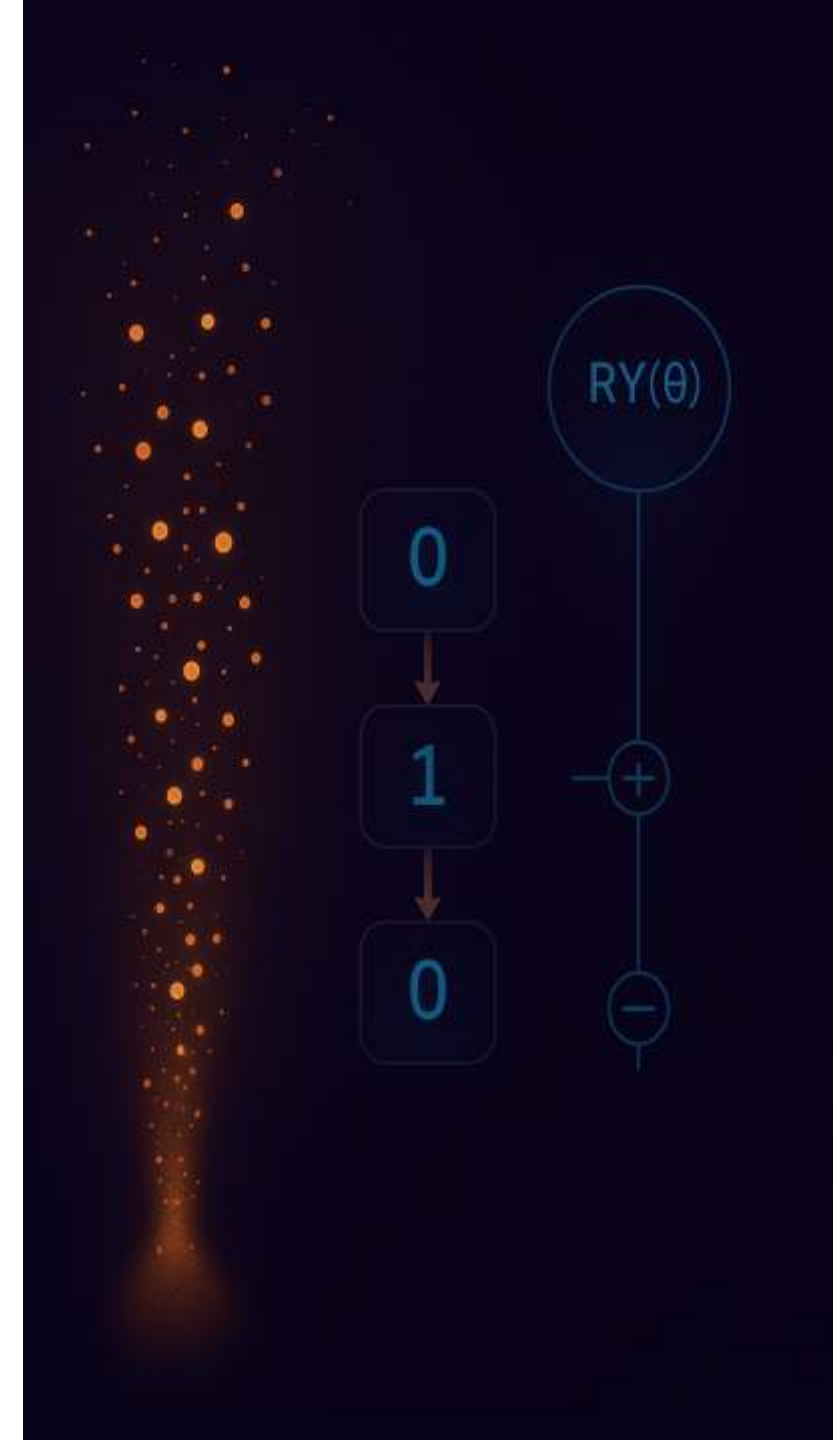
Initialize Source State

- Source : 56,128,200
- Amplitude = encode concentrations to position register
- Distribution to velocity states using $\sqrt{w}$ factors and normalize

Collision Step (Diffusion and Drift)

- Apply velocity-space unitary

$$U_c = R(\theta)P(\phi)$$



- $R(\theta)=0.30$: mixes right or left velocity states, controls diffusion rate
- Drift Bias is introduced by

$$P(\phi) = \text{diag}(1, e^{i\phi}, e^{-i\phi}, 1) \text{ , here } \phi = 0.08$$

Streaming Step

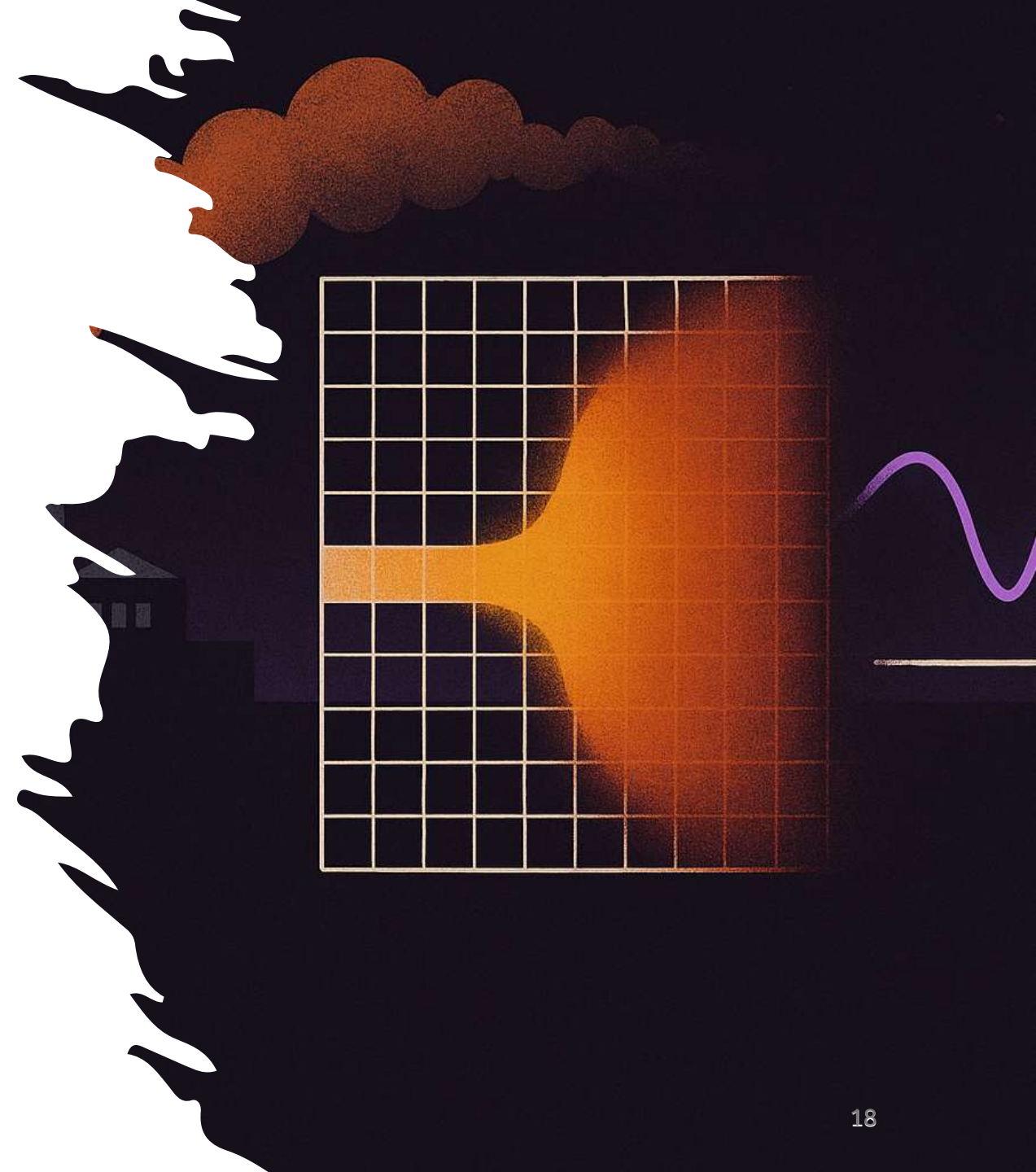
- Carried by position update either right or left, $x \pm 1$.

Hybrid Decay

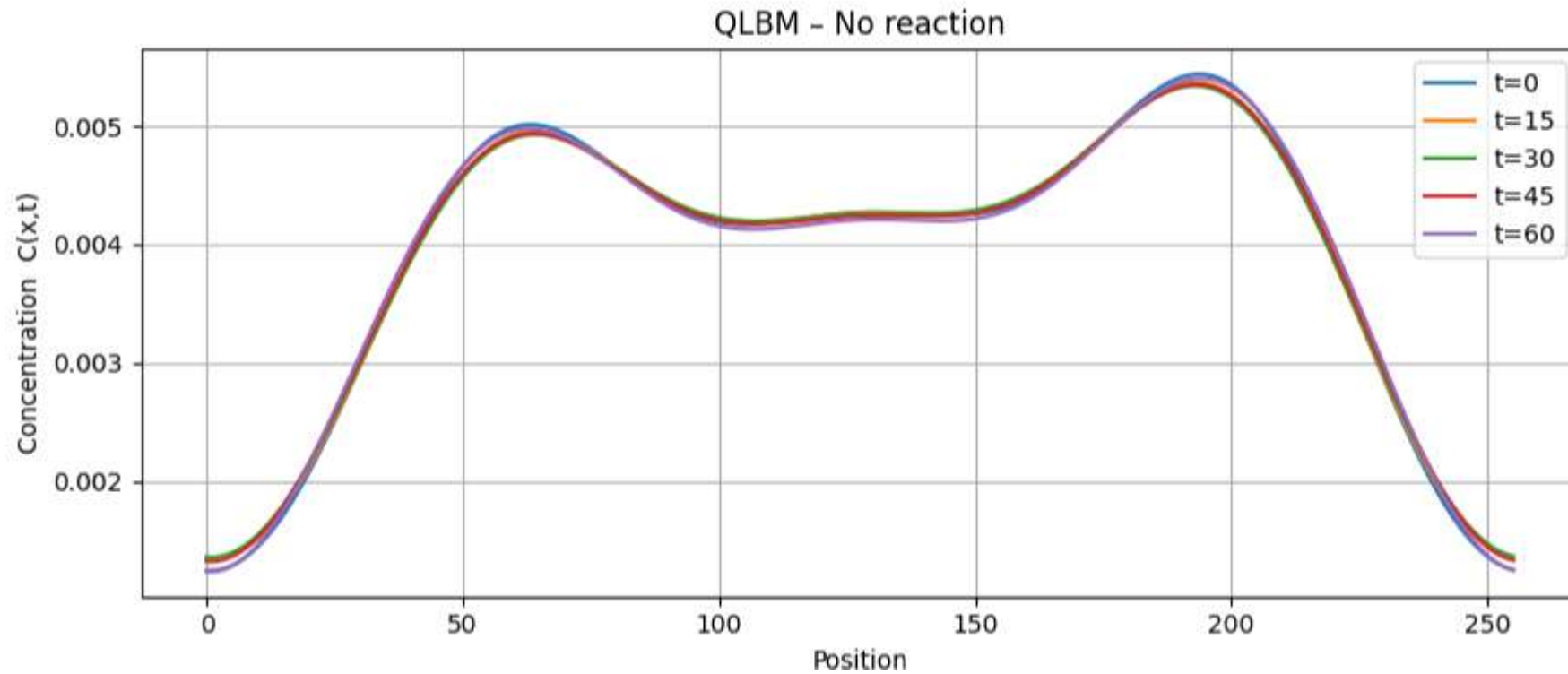
- The total mass(classically)

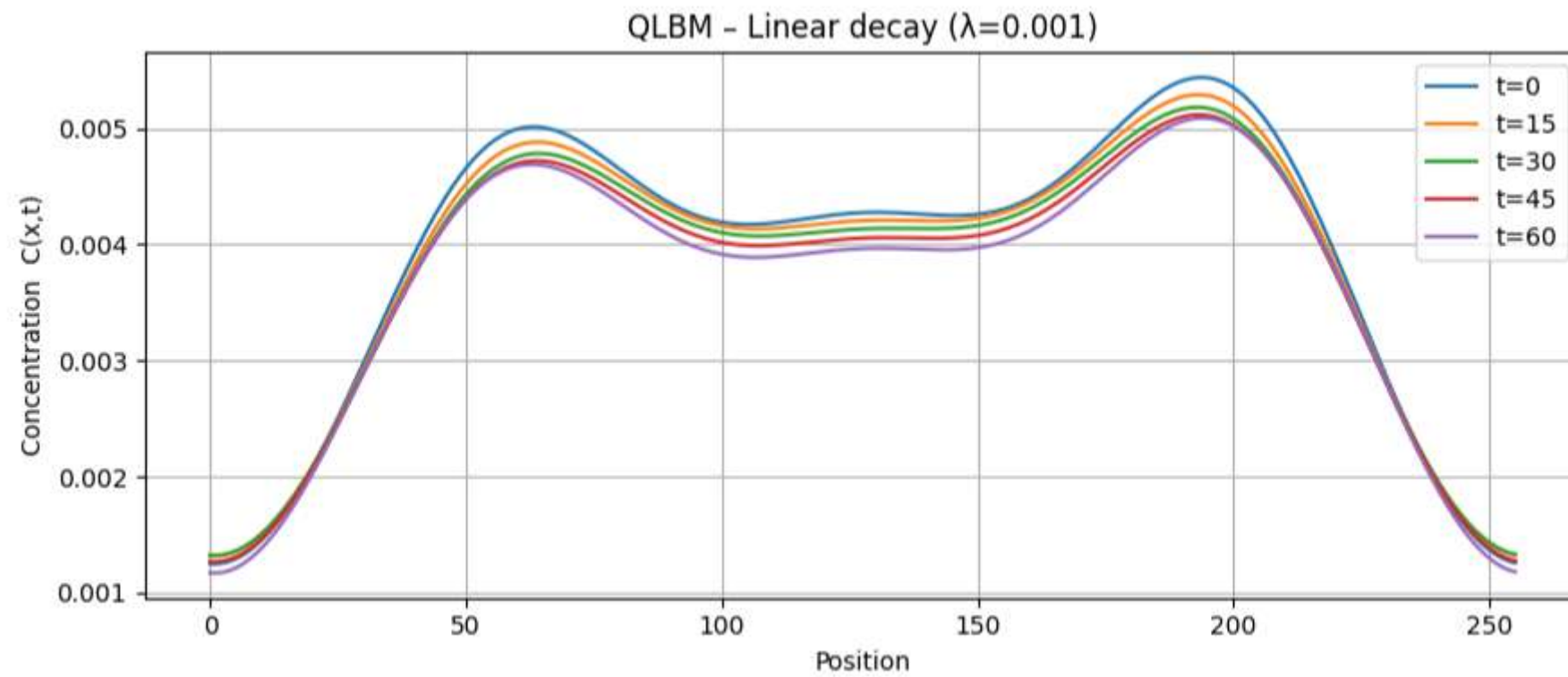
$$M_{t+1} = M_t(1 - \lambda) - kM_t^2 \sum_x p(x, t)^2$$

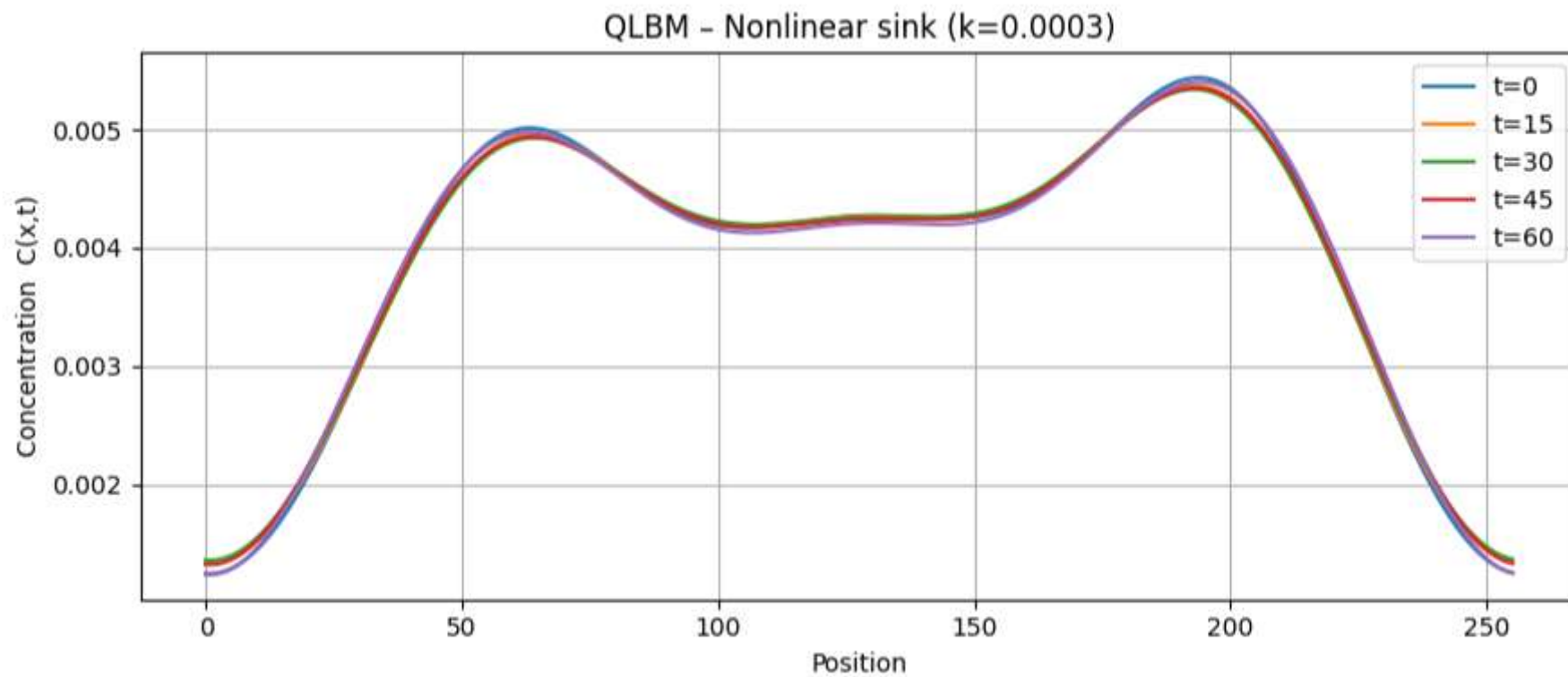
Here, $\lambda = 0.001$, $k = 3 \times 10^{-4}$

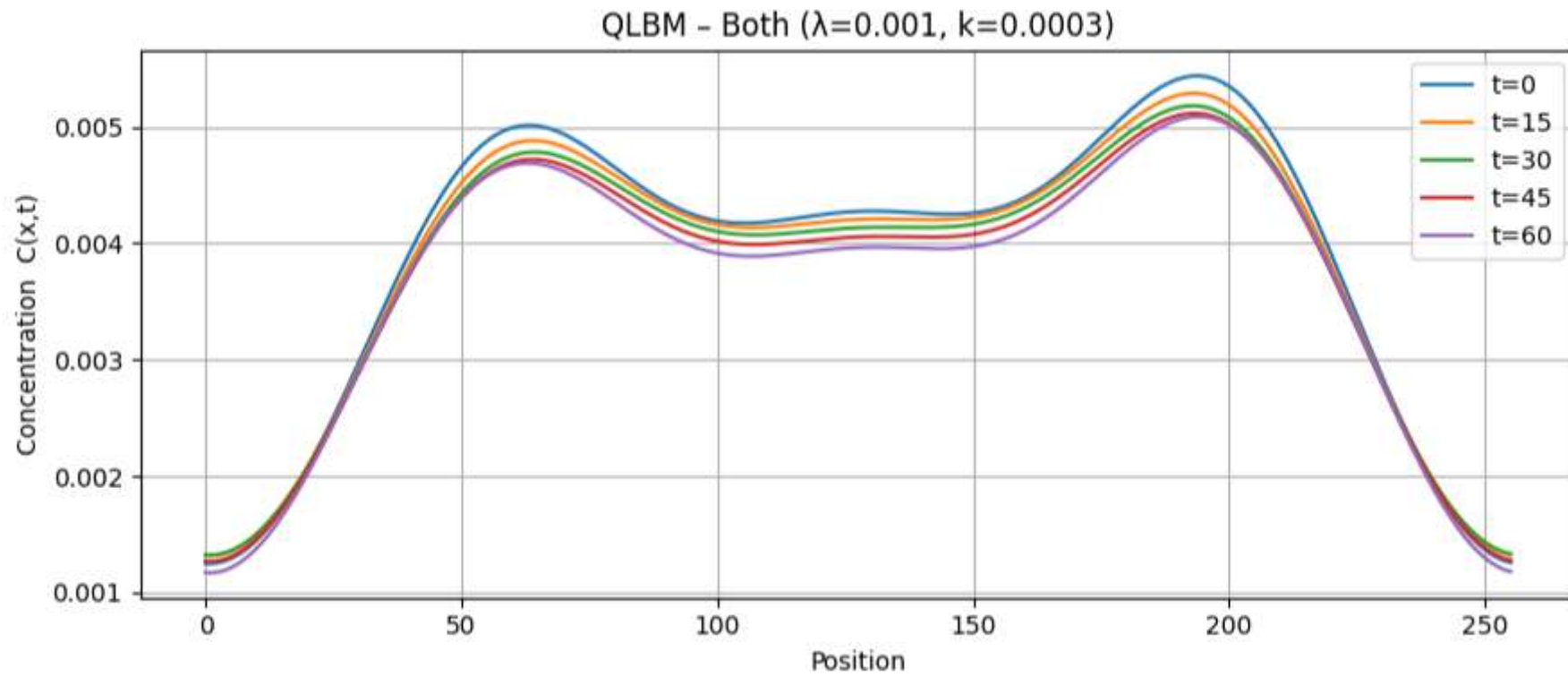


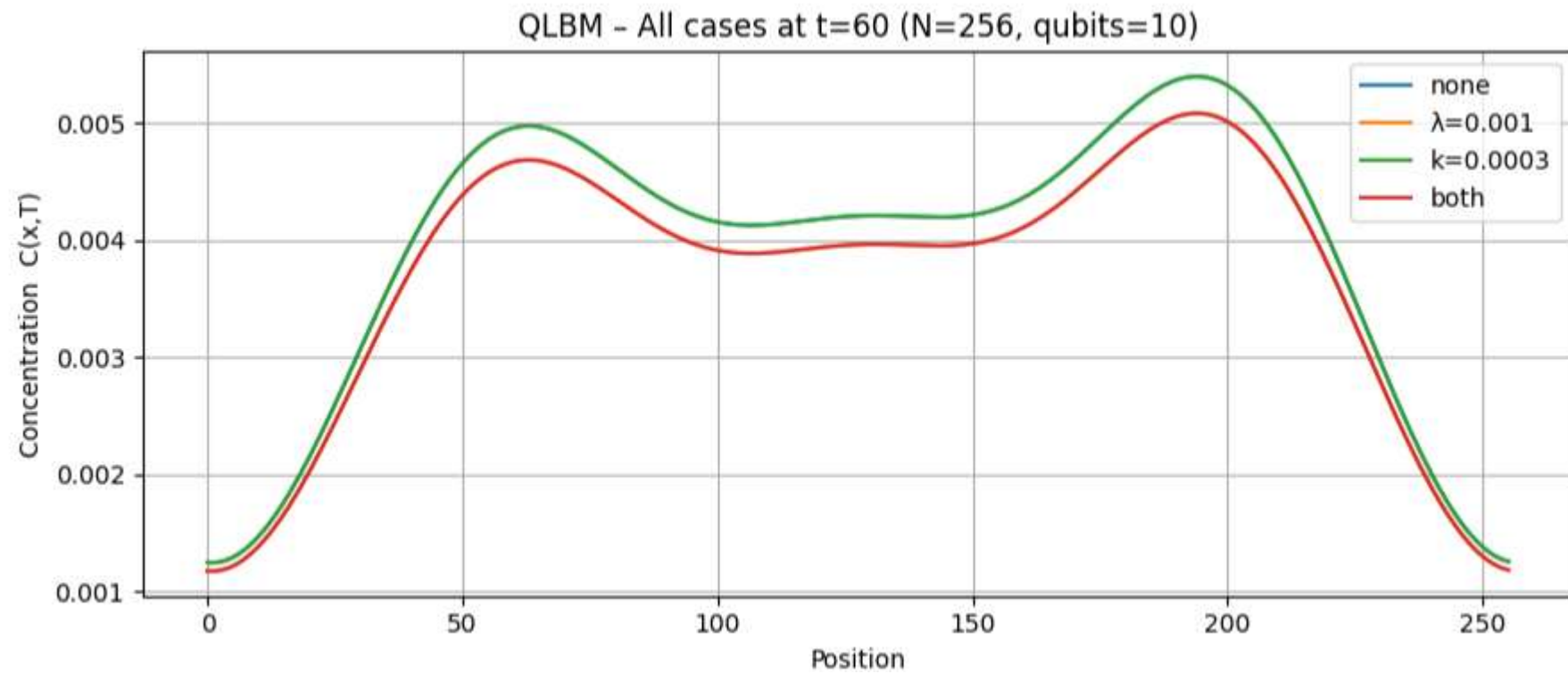
Results - QLBM



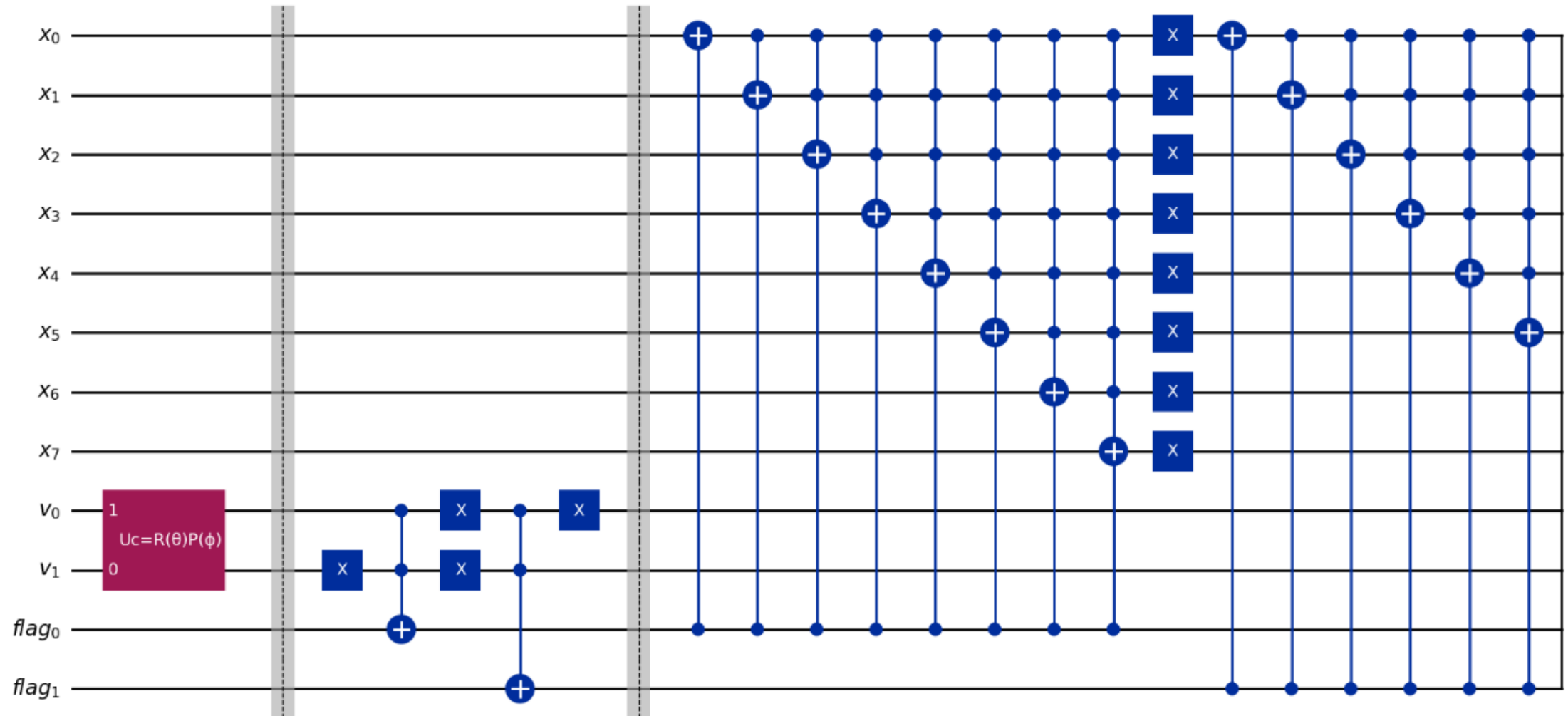






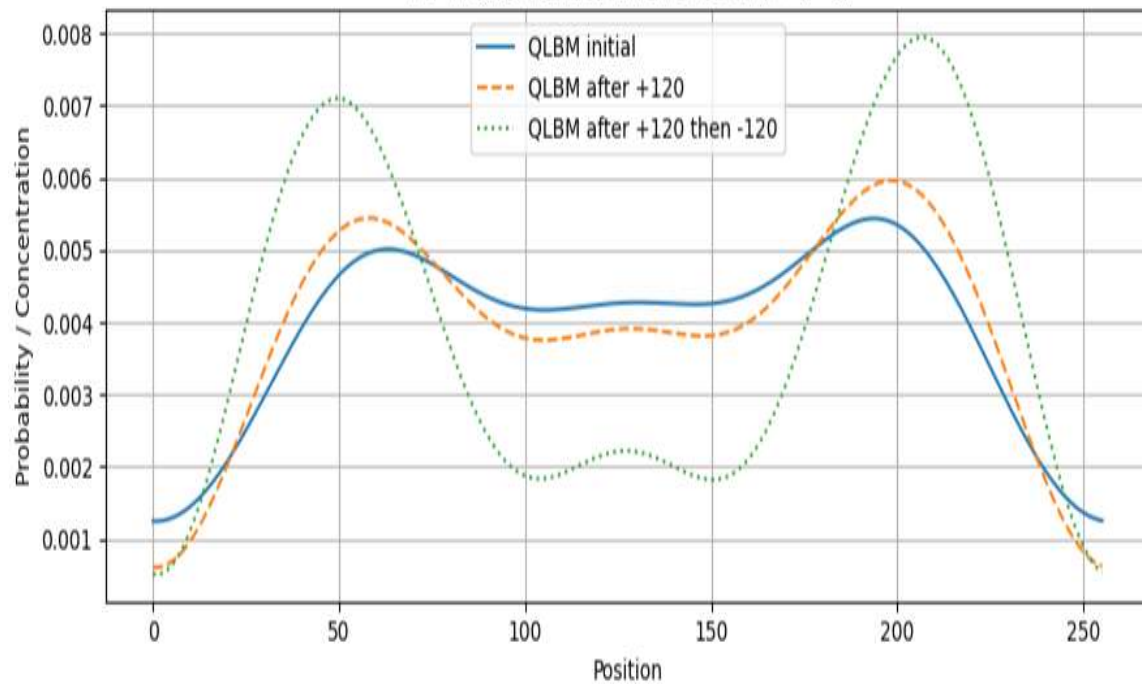


One Step QLBM Circuit

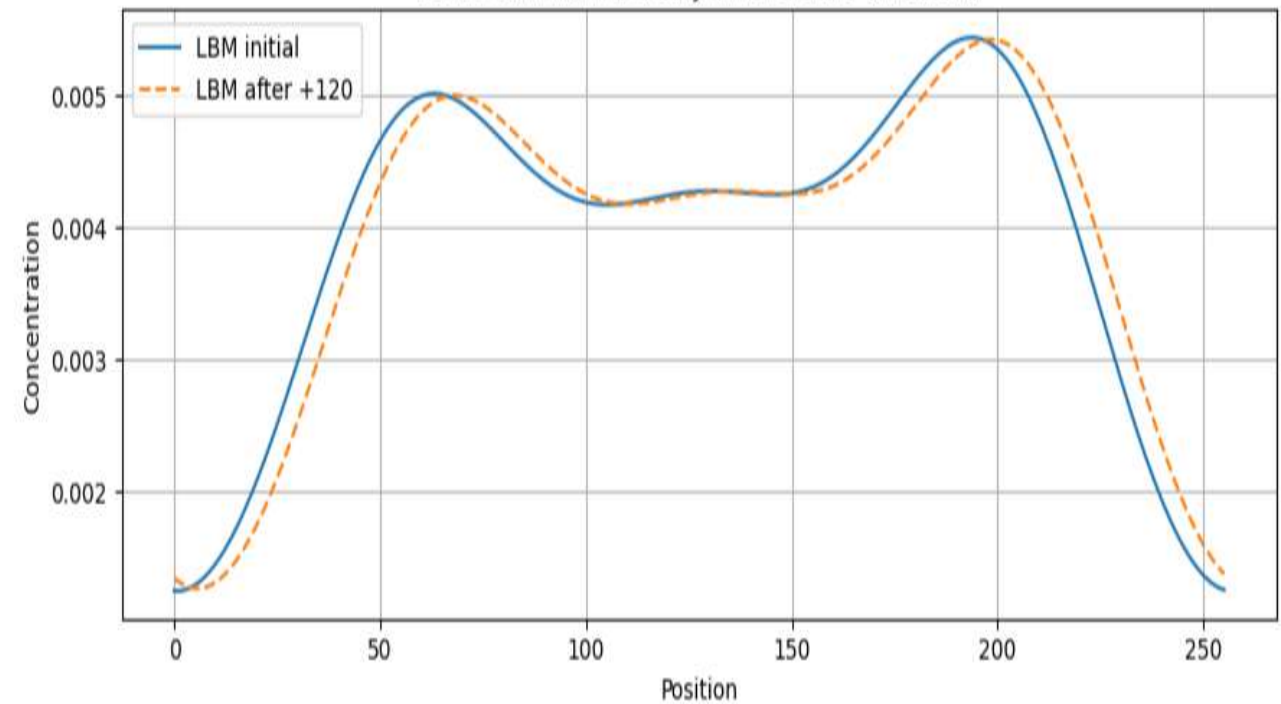


Time Reversal Capability

EXP1: QLBM exact time-reversal ($\lambda=k=0$)



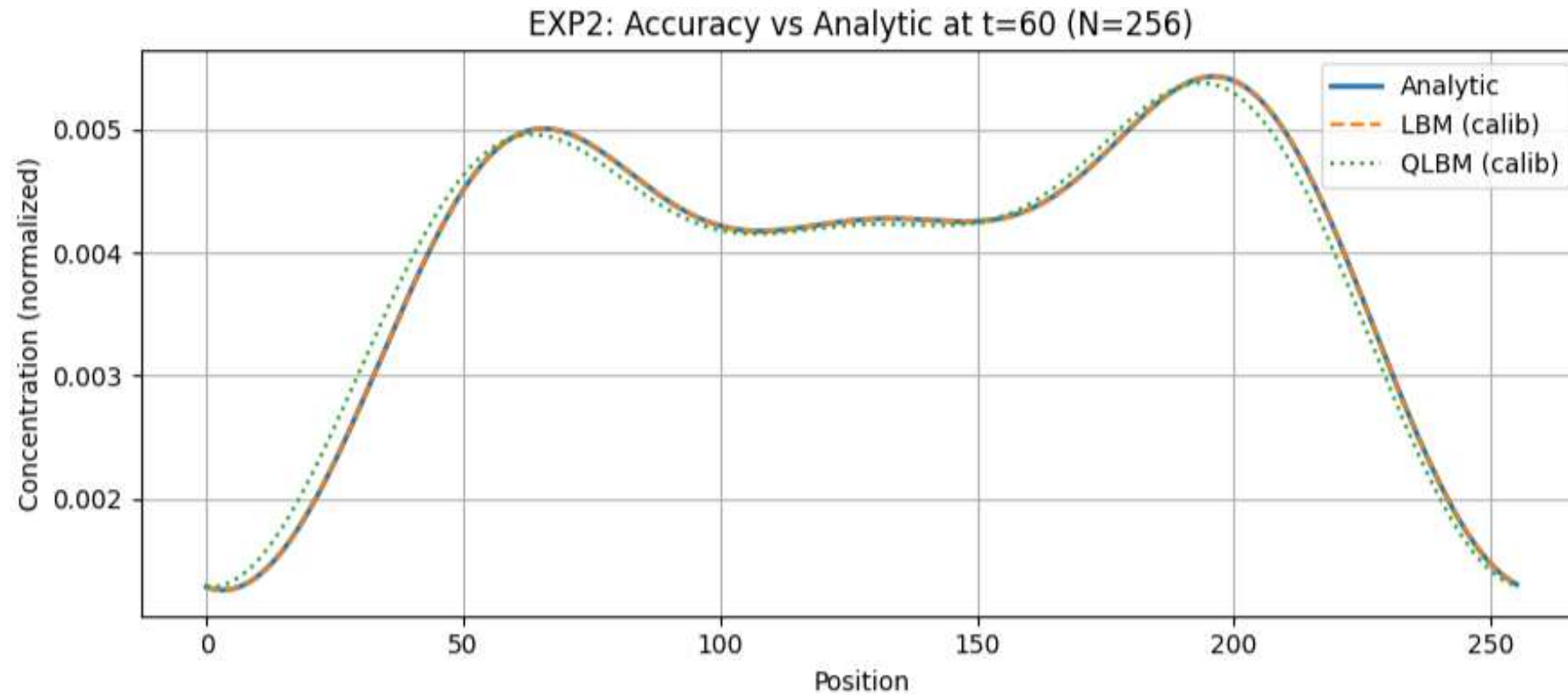
EXP1: LBM forward only (irreversible diffusion)



Accuracy vs Analytic Solution

Setup:

- QLBM $\theta = 0.340$
- LBM $D = 0.028$, Compare at $t=15,30,45,60$



Conclusion

- Implemented advection diffusion with multiple gaussian pollutant sources
- QLBM preserved the initial profiles better under reversal and showed reduced numerical diffusion
- QLBM maintained amplitude and symmetry , both LBM and QLBM matched for short time solutions

Overall Impression

- Classical LBM is efficient, scalable and established
- QLBM is promising for high precision, low diffusion and reversible fluid dynamics, which is relevant for the applications like pollution tracking

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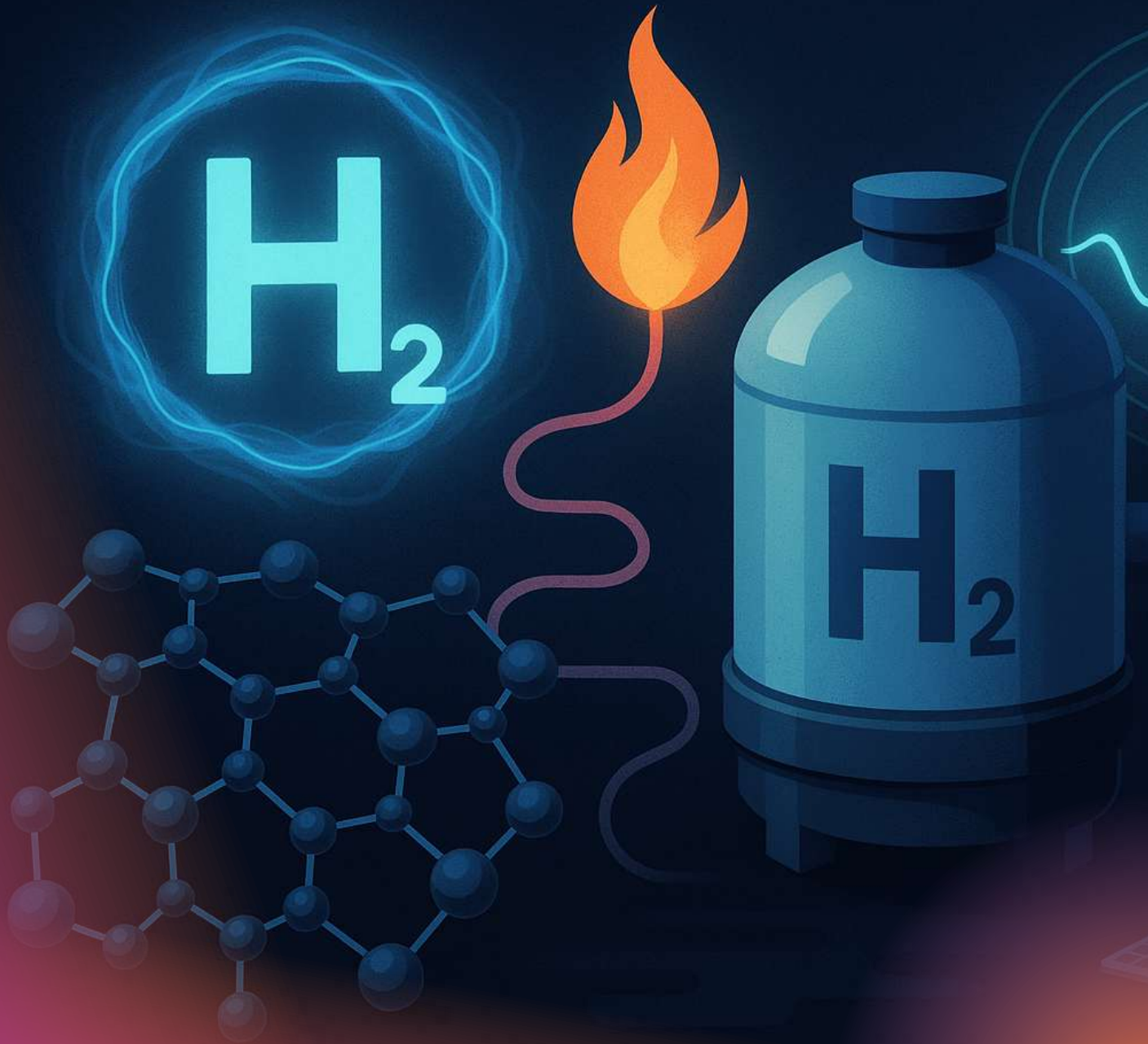
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<https://bayanbox.ir/view/2475672740784659029/Lattice-Boltzmann-Method-Fundamentals-and-Engineering-Applications-with-Computer-Codes.pdf>

Thank you

Q & A

Contact - LinkedIn



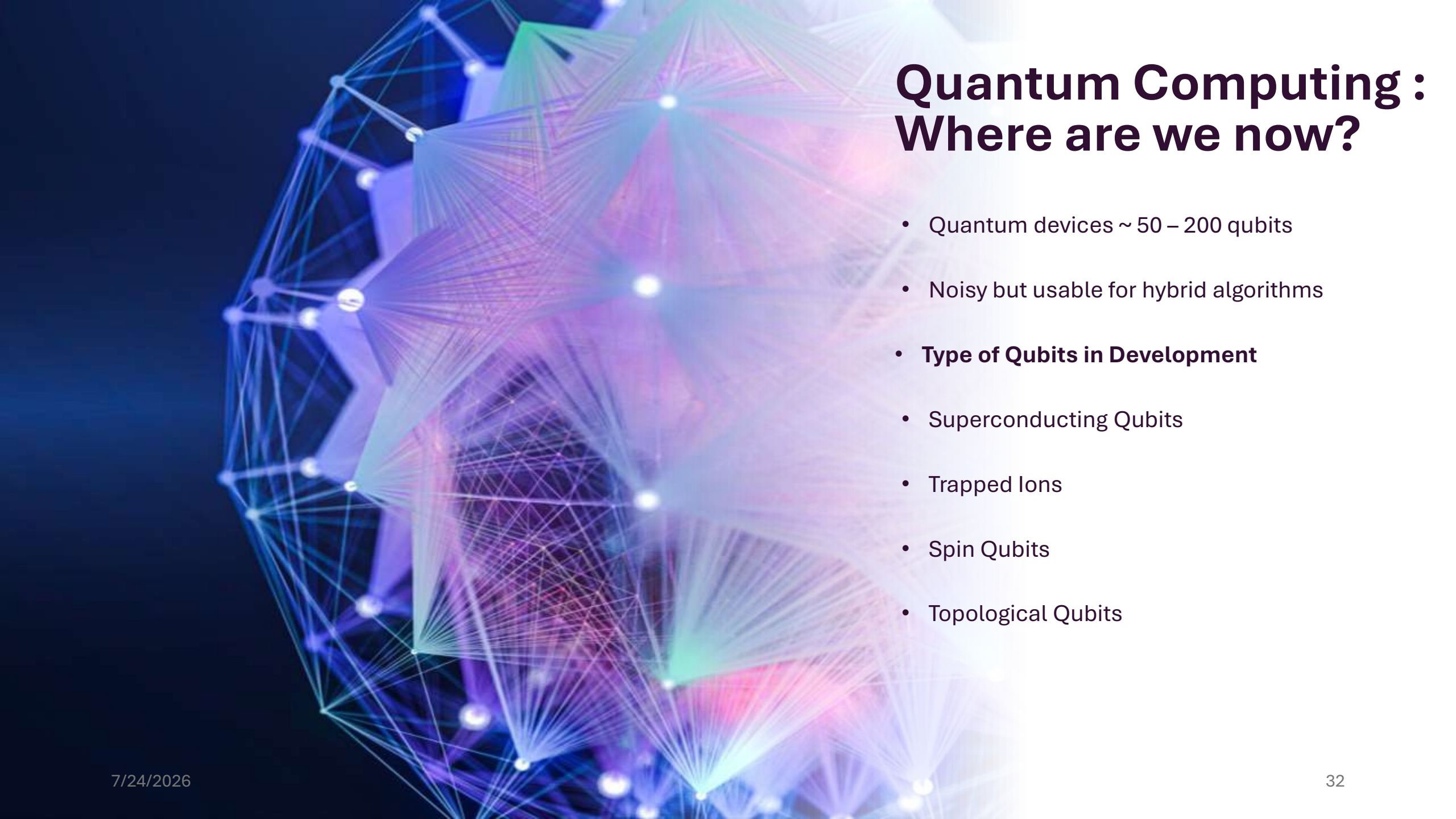


Quantum-AI for Energy Optimization

Problem: Hydrogen production is energy-intensive and must align with variable renewable supply

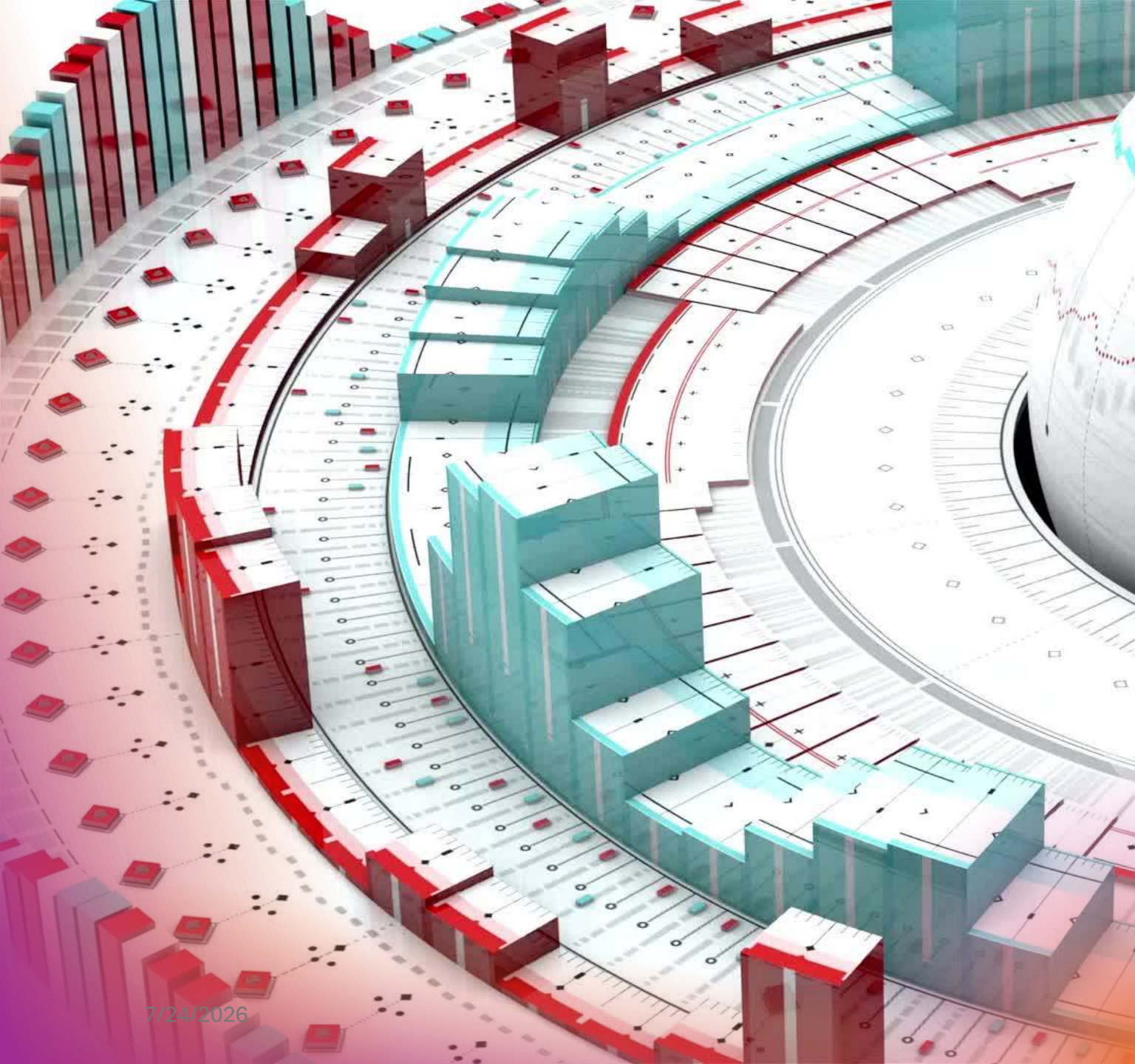
Constraints:

- Renewable availability (solar, wind)
- Electrolyzer capacity
- Storage limitations
- Electricity price variation



Quantum Computing : Where are we now?

- Quantum devices ~ 50 – 200 qubits
- Noisy but usable for hybrid algorithms
- **Type of Qubits in Development**
 - Superconducting Qubits
 - Trapped Ions
 - Spin Qubits
 - Topological Qubits



Real-World Examples & Industry Collaborations

Companies actively implementing quantum methods:

- Mercedes-Benz & IBM: Quantum simulations for hydrogen storage materials.
- Toyota & Hyundai: Quantum-enhanced catalyst design for fuel cells.
- BP & IBM: Quantum optimization in hydrogen production and storage.

Current Challenges & Future Directions

- **Quantum Hardware Limitations:**
Need for more reliable, scalable quantum processors.
- **Integration Complexity:**
Challenges combining quantum methods into classical workflows.
- **Industry Readiness & Adoption:**
Bridging the gap from proof-of-concept to full-scale deployment.

CFD: Classical Roots

Simulates fluid flow using numerical solutions to Navier–Stokes equations

Applications

- Aerospace → airflow over wings
- Automotive → engine cooling, aerodynamics
- Biomedical → blood flow, drug dispersion
- Energy → combustion, hydrogen flow
- Climate → weather patterns, atmospheric modeling



What Are Quantum Technologies?

- Built on quantum principles like superposition, entanglement, and tunneling.
- Represent the **second** quantum revolution - engineering quantum systems

Key Areas of Quantum Technology

- Quantum Sensing
- Quantum Communication
- Quantum Computing